

Lab Problem 3

Objective:

To implement Simple Linear Regression from scratch using the Normal Equation approach and verify the optimized parameters (θ) against a standard dataset.

Dataset Used:

- **Name:** Synthetic Linear Trend
- **Generation:** You must generate your own training data with 1 feature (x)
- **Step A:** Generate 20 training samples ($N = 20$)
- **Step B:** Generate(x) values uniformly spaced between 0 and 2.
- **Step C:** Generate (y) values using the equation:
$$y_{true} = 3x + 4$$
- Then add Gaussian noise with a standard deviation of 0.1.

Task:

You must perform the following steps in order:

Task 1: Generate Data

- Generate the feature vector (x) and target vector (y) as described above.

Task 2: Model Implementation (From Scratch)

- Implement a function `solve_normal_equation(x, y)` that performs the following:
 1. **Bias Term:** Transform the input (x) into a design matrix (X) by adding a column of ones ($x_0 = 1$)
 2. **Matrix Math:** Calculate the optimal parameters (θ) using the closed-form formula:
$$\theta = (X^T X)^{(-1)} X^T y$$
 3. **Return:** The intercept (θ_0) and slope (θ_1)

Task 3: Evaluation & Plotting

- Train a `sklearn.linear_model.LinearRegression` model on the same data for verification.
- **Plot:** Create a scatter plot of the training data. Overlay two lines:
 1. The line generated by your Normal Equation θ .
 2. The line generated by Sklearn.

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4. Deliverables:

Parameter Table: Compare the values found by your implementation vs. Sklearn.

Parameter	True Value	Your Normal Eq. Value	Sklearn Value
Intercept (θ_0)	4.0	...	...
Slope (θ_1)	3.0	...	...